2 Lines in the plane

Sheet 5 Mon 2 Nov. 2026 *Direction vector of a line*

1. Warm-up · Quick Give a vector collinear to $\vec{u}\begin{pmatrix}2\\-4\end{pmatrix}$ with simpler integer coordinates.
2. Core · Method Give a direction vector of (AB) for $A(1,2)$ and $B(4,8)$.
3. Core + · MCQ A direction vector of the line through $A(0,1)$ and $B(2,-3)$ is:

   A. $\begin{pmatrix}1\\-2\end{pmatrix}$ **B.** $\begin{pmatrix}1\\-4\end{pmatrix}$ **C.** $\begin{pmatrix}-4\\-2\end{pmatrix}$ **D.** $\begin{pmatrix}2\\4\end{pmatrix}$
4. Review · Quick Marks: 8, 12, 12, 15, 18. Calculate the mean, the median and the range.
5. Thread · Property ABC is right-angled at A and I is the midpoint of $[BC]$. What can be said about IA, IB and IC? Why?

Sheet 6 Tue 3 Nov. 2026 *Cartesian equation of a line*

1. Warm-up · Quick Calculate $3x-2y+1$ for $x=1$ and $y=-2$.
2. Core · Method The line d has equation $2x+3y-6=0$. Give a direction vector and a point of d.
3. Core + · True/false True or false? Justify. The lines $2x+3y-6=0$ and $4x+6y+1=0$ are parallel.
4. Review · Table Mode of transport of 30 students.

Transport	Bus	Bike	Walking	Car
Frequency	12	5	8	5

   Give the proportion of students who come by bus and the angle of the "bike" sector in a pie chart.
5. Thread · Quick On a number line, A has coordinate -2 and B has coordinate 3. Give AB and the coordinate of the midpoint of $[AB]$.

Sheet 7 Thu 5 Nov. 2026 *Slope-intercept equation and gradient of a non-vertical line*

1. Warm-up · Quick Make y the subject of $2x+y-5=0$.
2. Core · Method Give the gradient and the y-intercept of the line $3x-2y+4=0$.
3. Core + · MCQ The gradient of the line (AB) with $A(1,2)$ and $B(5,0)$ is:

   A. 2 **B.** $\frac{1}{2}$ **C.** -2 **D.** $-\frac{1}{2}$
4. Review · Quick Calculate $\frac{3}{4}+\frac{5}{6}$.
5. Thread · Problem Two gears have 12 and 18 teeth. After how many turns of each do they return together to their starting position?

Sheet 8 Mon 9 Nov. 2026 *Drawing a line given by an equation*

1. Warm-up · Quick For $y=-2x+3$, calculate y when $x=0$ and then when $x=2$.
2. Core · Method Give two points that allow you to draw the line $2x+y-4=0$.
3. Core + · Quick What is the line with equation $x=3$? Give a direction vector. Does it have a gradient?
4. Review · Quick Compare $\frac{3}{7}$ and $\frac{4}{9}$.
5. Thread · Quick Solve in $\mathbb{R}$: $5x-7=2x+8$.

Sheet 9 Tue 10 Nov. 2026 *Finding the equation of a line: two points, a point and a direction vector, a point and a gradient*

1. Warm-up · Quick Gradient of the line through $(1,3)$ and $(3,7)$?
2. Core · Method Slope-intercept equation of the line with gradient 2 through $A(1,3)$.
3. Core + · Method Line through $A(1,2)$ with direction vector $\vec{u}\begin{pmatrix}2\\-1\end{pmatrix}$: Cartesian equation using the determinant of $\overrightarrow{AM}$ and $\vec{u}$.
4. Review · Quick Simplify $2^5 \times 2^{-3}$.
5. Thread · Table → formula An affine function f satisfies: $f(0) = -1$, $f(1) = 1$, $f(2) = 3$. Give $f(x)$.

Sheet 10 Thu 12 Nov. 2026 *Lines and affine functions: linking the representations*

1. Warm-up · Quick $f(x) = -3x + 4$. Calculate $f(2)$ and the preimage of 10.
2. Core · Method Find the affine function f such that $f(1) = 1$ and $f(3) = 5$.
3. Core + · True/false True or false? Justify. Every line in the plane is the graph of an affine function.
4. Review · Quick A fair 6-sided die is rolled. What is the probability of getting a multiple of 3?
5. Thread · Quick ABC is right-angled at A with $AB = 6$ cm and $AC = 8$ cm. Calculate BC.

Sheet 11 Mon 16 Nov. 2026 *Collinearity, parallelism, intersection of lines*

1. Warm-up · Quick Does the point $(3,5)$ lie on the line $y = 2x - 1$?
2. Core · Method Relative positions of $d:\ y = 3x + 1$ with $d':\ y = 3x - 2$, then with $d'':\ y = -x + 1$.
3. Core + · Method Show in two ways that $A(1,4)$, $B(2,7)$ and $C(4,13)$ are collinear.
4. Review · Quick On a map with scale $\dfrac{1}{50{,}000}$, two towns are 4 cm apart. What is the real distance?
5. Thread · Method Programme: choose a number x; add 3; multiply by 2; subtract 6. Calculate the result for $x = 5$, then write an expression in x and simplify it. What do you notice?

Sheet 12 Tue 17 Nov. 2026 *Intersection of two lines; Python collinearity function*

1. Warm-up · Quick Solve $2x + 1 = -x + 7$.
2. Core · Method Coordinates of the intersection of $y = 2x + 1$ and $y = -x + 7$.
3. Core + · Code reading What do `alignes(0,0,1,2,2,4)` and `alignes(0,0,1,2,2,5)` return?

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == 0
```

4. Review · Expand Expand and simplify $3(x - 2)$.
5. Thread · Prove Show that the sum of three consecutive integers is a multiple of 3.